Can Question Validation Criteria Improve the Quality of LLM-Generated Multiple-Choice Questions?

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***Abstract*—Adaptive assessment has been extensively researched as a key strategy for providing formative assessments in education. Fundamental to adaptive assessment is the development and ongoing maintenance of a substantial and well-structured question bank. However, manual question development by human instructors is time-consuming and poses a significant barrier to adopting adaptive assessment. To address this problem, research has been conducted in the use of Large Language Models (LLMs) for automated question generation. However, LLMs tend to hallucinate, necessitating human-in-the-loop question validation. Furthermore, to ensure alignment with the instructional standards and learning outcomes of a course, it is essential that instructors verify that the generated questions satisfy a defined set of Question Validation Criteria (QVC). In this study, we evaluate the effectiveness of embedding QVC directly into LLM prompts for the automatic generation of multiple-choice questions (MCQ) with feedback for use in adaptive formative assessments. Using a novel web-based application for question generation, MCQs were produced for three undergraduate-level courses. Validation by subject matter experts revealed no statistically significant improvement in quality between questions generated with and without embedded QVC in the prompts. These findings contribute to the growing body of research indicating that prompt engineering alone may be insufficient to achieve improvements in the quality of generated output for educational purposes.**

***Index Terms*—Automatic Question Generation, Generative AI, Large Language Models, Formative Assessment, Adaptive Assessment**

I. Introduction

Online learning has emerged as a transformative educational paradigm, enabling the realization of adaptive learning systems (ALS). Within this framework, personalized learning paths [1], [2], student learner models [3], content [4], activities [5], [6], and feedback [7], [8] can be provided on-demand to better support the needs of individual learners. The emphasis on individualization in adaptive assessment makes it ideal for providing formative assessments, aimed at identifying learners' abilities and providing formative feedback to help guide their learning. During an adaptive formative assessment, an adaptive algorithm selects the optimal sequence of questions to present to learners based on their responses. The algorithm tries to minimize the number of questions required to accurately estimate their mastery on a set of learning outcomes (LOs). Because the sequence of questions is adaptive, and not set by the instructor, adaptive assessments require a question bank containing a sufficient number of questions corresponding to the LOs covered by the assessment so that questions are not repeated. The process of manually creating these questions is time consuming and costly for instructors and educational institutions. To reduce instructors' workload, extensive research has been done on how questions can be generated automatically [9].

The recent advancement of large language models (LLMs), such as GPT-3 and GPT-4, has shown tremendous capabilities in various tasks, including automatically generating questions with appropriate prompts and parameters [10], [11]. These LLMs offer immense opportunity and pose many open challenges which can limit their adoption in educational settings [12]. An example of this is that hallucinations produced by LLMs often resemble routine human writing, making them difficult to detect and, in many cases, amount to misinformation [13]. As a result, incorporating a human-in-the-loop procedure, where instructors validate the outputs of automatic question generation (AQG) using LLMs, is essential to ensure quality and reliability before the questions are presented to learners [14]–[17]. However, the process of question validation may impose an increased workload on instructors if a large number of questions are generated. Therefore, methods for ensuring the quality of questions developed by AQG using LLMs are needed.

Multiple-choice questions (MCQs) are the most commonly employed question type in adaptive assessment systems. MCQs offer several advantages over open-ended question types, including greater time efficiency, ease of automated

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grading, and objective scoring [18]. Additionally, if designed well, MCQs provide the ability to test a range of learners' cognitive skills [19]. Therefore, research has extensively explored how LLM techniques can be employed to generate high-quality MCQs [20]. However, there is a lack of research specifically focusing on how question validation criteria (QVC) can be used in the context of AQG using LLMs to generate MCQs.

In this study we seek to answer the following research questions: a) what QVC can instructors use to validate MCQs generated using LLMs, and b) will they improve the question quality if embedded in the prompt. To answer these questions, a set of QVC was developed by synthesizing and adapting MCQ writing guidelines found in existing literature. These QVC were then used to generate MCQs for three undergraduate level courses. Two sets of questions were generated for each course: with and without QVC in the prompt. A comparison study to evaluate the quality difference between the two sets of questions for each course was conducted with subject matter experts providing ratings. The results indicate that embedding QVC in the prompt has varying effects across disciplines but, overall, yields similar outcomes to prompts without QVC.

II. Related Work

A. Guidelines for Developing Effective Multiple-Choice Questions

Developing effective MCQs is a nuanced process that requires careful consideration of various educational theories and principles to ensure valid, reliable, and effective assessments. Several sets of guidelines for constructing MCQs have been developed. Haladyna & Downing [21] created a taxonomy of multiple-choice item-writing rules. It includes 43 suggestions from general item writing, stem construction, option development, correct option development, and distractor development. Tarrant et al. [22] investigated around 20 frequent item writing flaws in MCQs.

Dibattista [23] presents strategies for creating MCQs in higher education that assess higher-order cognitive skills such as understanding, applying, analyzing, and evaluating. Dibattista emphasizes the need for MCQs to go beyond simple recall, pushing educators to construct questions that engage deeper levels of cognitive processing. Gottlieb et al. [18] provide a comprehensive guide to educators on crafting high-quality MCQs. The authors outline ten strategies for question creation, including determining purpose, assembling a writing team, writing succinctly, and assessing knowledge and comprehension.

These guidelines have provided valuable insights into how to create MCQs from a variety of perspectives, including relevance and alignment, clarity and comprehension, fairness and bias, plausibility of distractors, readability of choices, test-wise and logical cues, independence of choices, and feedback quality.

B. Techniques for Automatic Question Generation

In the hope of expediting the creation of questions, researchers have long been studying AQG techniques [24]–[28]. AQG leverages advancements in artificial intelligence and natural language processing (NLP) to create questions dynamically based on input texts or specific LOs. AQG systems aim to support educational environments by providing scalable, efficient, and personalized question generation tools that can increase the efficiency of educators.

Several recent reviews highlight the important advances in the field of AQG in recent years. Kurdi et al [29] provide a systematic review of the field of AQG up to 2019. They describe the success and difficulties faced by the creation and adoption of AQG systems. Das et al [27] performed a survey on AQG techniques used to generate questions and evaluate their answers. Mulla et al [25] performed a review of AQG techniques focusing on methodologies, available datasets, challenges, and applications. They review recent advances in the use of transformer architectures and language models, including GPT-2 and BERT for AQG.

Bachiri and Mouncif [30] outline a system architecture that uses machine learning (ML) techniques and NLP to generate questions, 20% of which did not pass human evaluation. Bachiri and Mouncif's [30] work provides essential insight into how crucial humans are in evaluating LLM content generation for education. Similar to this work, Akyön et al. [31] and Chomphooyod et al. [32] also acknowledge the importance of high-quality exam-style questions in education and the challenges associated with manual creation. Wen et al. explored the use of knowledge graphs (KGs), SPARQL, and reinforcement learning [33] for question generation. Other researchers have also explored using KGs for QA systems [34]. Recently, Chen and Shiu [35] introduced a KG and Retrieval Augmented Generation (RAG) [36] based approach called knowledge augmented question generation to control the difficulty of generated questions.

C. Generating Questions using Validation Criteria

Validation is an essential component of AQG. Gorgun and Bulut [37] developed a comprehensive survey on quality criteria and evaluation in AQG. Raina and Gales [38] investigated MCQ generation for personalized English comprehension assessments. They proposed multiple performance criteria, such as grammatical correctness, answerability, diversity, and complexity. The framework uses input context paragraphs to provide the grounding need to generate relevant English comprehension assessments [38].

Maity et al. [39] investigated the ability of GPT-4 Turbo to generate educational questions that align with Bloom's Revised Taxonomy. Utilizing a zero-shot learning approach, they prompt GPT-4 Turbo to generate questions from textbooks across various subjects, assessing the alignment of these questions with different cognitive levels of Bloom's taxonomy using ML models and human expert validation. Their findings [39] revealed that GPT-4 Turbo is particularly effective in generating questions at the "understanding" level but faces

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challenges in accurately distinguishing between higher-order cognitive levels such as "analyzing" and "evaluating." The study highlights the need for further refinement of GPT-4 Turbo's question generation capabilities to better meet educational standards, particularly in crafting high-quality questions at more complex cognitive levels.

III. Methodology

Building on prior research, this study employed a structured methodology to evaluate whether embedding QVC into LLM prompts enhances the quality of automatically generated MCQs. The methodology encompassed the development of a QVC framework, the implementation of two prompting strategies for LLM-based MCQ generation, and the deployment of a web-based system to support question creation and instructor-led validation.

A. Proposed Question Validation Criteria

The framework used to develop the QVC for this study is primarily based on the 10 strategies for question creation provided by Gottlieb et al. [18]. By synthesizing the strategies and guidelines for creating and evaluating MCQs advocated in this research, the following five requirements were identified:

1) Alignment: Since MCQs are used for assessing an existing curriculum, questions should match the category or domain of the intended goal or learning outcomes [18]. Bloom's revised taxonomy for cognitive skills (remember, understand, apply, analyze, evaluate, create) can be used for specifying learning outcomes to align MCQs with. MCQs can then be designed to test various levels of cognitive skills [19], [40].
2) Fairness: The design and scoring of MCQs should follow clear strategies to ensure fairness. Five axioms have been suggested to guide the development of effective scoring strategies, supporting the theory that MCQs, when designed with specific criteria in mind, can serve as a powerful tool for educational assessments [41].
3) Effectiveness: Assessment questions need to possess high discrimination ability to effectively differentiate among learners with varying knowledge or skill levels. In other words, these questions should be designed to effectively identify and reflect the different understanding and mastery levels across a broad spectrum of learner capabilities [42]. Questions with such effectiveness allow educators to pinpoint learners' strengths and weaknesses, guiding targeted intervention and support where necessary.
4) Reliability: MCQs should be well-constructed to reliably measure learner knowledge. They should be free of guessing and effective in detecting misconceptions.
5) Instructive feedback: Instructive feedback plays a vital role in formative assessment by clarifying why an answer is correct or incorrect and offering actionable guidance to help learners improve their understanding and skills. To support this, each MCQ in the study included not only the question and answer options but also option-level feedback. This feedback is designed to provide clear, informative, and pedagogically meaningful explanations for each option, helping learners at different stages of understanding engage more deeply with the content and correct misconceptions. Generating adaptive feedback has been previously explored with bandit algorithms [43].

TABLE I
Multiple Choice Question Design Requirements

No	Requirement	Aspect	Item
1	Alignment	Relevance and Alignment	The question should align with the learning outcome and its cognitive level.
2	Fairness	Fairness and Bias	The question should be free of race, ethnic and sex bias.
3	Effectiveness	Clarity and Comprehension	The question should be clear, concise, complete, and easy to understand.
4		Plausibility of Distractors	The choices should include plausible and homogeneous distractors with parallel length.
5		Readability of choices	The choices should be in logical, alphabetical, chronological, or numerical order.
6	Reliability	Test-wise and logic cues	The choices should be similar in length, use the same expression style, and have consistent grammar.
7		Independency of choices	The choices should be independent and free of overlapping responses and extreme modifiers.
8	Instructive Feedback	Feedback Quality	The feedback should provide a clear, informative, and instructive explanation.

Based on these requirements, 8 QVC were created, shown in Table I. These criteria suggest that a well-balanced assessment through MCQs requires a holistic approach, considering the construction of questions, the methods used to evaluate their quality, and the cognitive demands placed on learners.

B. AQG using LLMs

1) LLM Question Generation Pipeline: While it is relatively easy to generate questions using LLMs [44] and adaptive practicing systems using LLMs have been discussed in prior research [45]–[47], finding a reliable prompting method that ensures high-quality questions in a specific format can be challenging. Different prompting strategies yield significantly different results [48], [49].

In this study, LangChain, a popular framework designed to facilitate the development of LLM-based applications, is used to generate MCQs using LLMs [50]. LangChain is particularly well suited for processing and sending prompts to OpenAI's GPT models. Additionally, it provides interfaces to other popular LLMs, allowing the reuse of code for testing different models. LangChain Chat models with OpenAI's GPT-4o-mini model are used [11]. Chat models are used instead

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of LLMs directly because they support assigning distinct roles to conversation messages and distinguishing between AI-generated, human-generated, and system-generated messages. OpenAI's GPT-4o-mini model was selected for its balance of cost-efficiency and output quality [51].

After MCQs are generated using LangChain, the output, received from the LLM as a string, is parsed using LangChain's Structured Output Parsers. This enables the transformation of the string output into a useful format, such as JSON, that can be stored in a database or displayed on a screen. Furthermore, LangChain provides intuitive methods for processing and querying documents using RAG techniques.

2) Question Generation Methods: Two prompting techniques were developed to evaluate MCQ generation with QVC.

- **Method 1: Single Prompt Without QVC**
  In this method, the LLM is instructed to generate a specific number of questions relevant to a given course, unit, and LO, without any additional constraints. Any relevant context from available course materials is added to the prompt using a RAG pipeline. This generation method acts as our base case.
- **Method 2: Single Prompt With QVC**
  The LLM is given the same directions as in the previous method, as well as the list of QVC (see Fig. 1). The LLM is instructed to ensure that all generated questions meet these criteria.

3) Question Generation System: The architecture of the client-server-based question generation system is shown in Fig. 2. LLM functionality is provided by the LangChain JavaScript library, integrated into the server application. Data for course objects (courses, units, LOs), MCQs, and course materials, are stored in the MongoDB database. Additionally, MongoDB Vector Search is used to store the vector embeddings of course materials used for RAG.

A web-based interface (see Fig. 3) enables instructors to create and manage courses, course materials, and MCQs. To generate questions, instructors select a question generation method, specify the course, unit, and LO, determine the number of questions to generate, and set the number of options each MCQ must have. The course, unit, LO, and the relevant document embeddings are retrieved. This data is then injected into the prompt template corresponding to the selected generation method. The prompt, LLM, and output parser work together to generate questions, which are formatted as an array of JSON objects and returned to the front-end application. Instructors can then review the generated questions and either reject or save them during the validation process.

```
You are the instructor of a course on {course}. You are creating
a formative assessment for the unit {unit} to test your
students' understanding of the learning objective
{learningObjective}

Use the following context to aid in generating the content:

{context}

Provide {numOfQuestions} multiple-choice questions. Each
question should have {numOfOptions} options and only a single
correct answer.

The question and its options should align with the following
criteria. If they do not, adjust them.

1. Relevance and Alignment: The question aligns with the
learning outcome and its cognitive level
2. Clarity and Comprehension: The question is clear, concise,
complete, and easy to understand
3. Fairness and Bias: The question is free of race, ethnic,
and sex bias
4. Plausibility of Distractors: The options include plausible
and homogeneous distractors with parallel length
5. Readability of Options: The options are in logical,
alphabetical, chronological, or numerical order
6. Test-wise and Logic Cues: The options are similar in length,
use the same expression style, and have consistent grammar
7. Independency of Options: The options are independent and
free of overlapping responses and extreme modifiers

You have already generated the following questions:

{previous_questions}

DO NOT repeat questions that you have already created.

The output must have the following format:

{parser.get_format_instructions()}
```

Fig. 1. Single Prompt With QVC MCQ generation prompt.

IV. Experimental Setup

A. Course and Learning Outcome Selection

To evaluate the effectiveness of automated MCQ generation with QVC across diverse academic contexts, we conducted an experimental study involving three undergraduate courses from distinct disciplines:

- Principles of Biology I (Biology)
- Administrative Law of Canada (Law)
- Group and Organizational Communication (Communication)

From each course, one representative unit was chosen for question generation. The number of LOs selected per course varied: Communication included 3 LOs, Biology included 5, and Law included 6. This variation enabled a more comprehensive evaluation of the question generation methods across courses of differing content complexity and structure.

B. Question Generation

For each LO in each course, a total of ten MCQs were generated, five using each of the two generation methods described in Section III-B2. Each question included four answer options, with only one correct answer. Additionally, every option was accompanied by generated feedback intended for students based on their selection. In total, 50 questions were generated for Biology, 60 for Law, and 30 for Communication.

C. Validation Process

After the questions were generated, they were compiled into Excel spreadsheets and emailed to the respective course instructors for evaluation. Instructors were asked to rate the overall quality of each question using a 5-point Likert scale, with 1 indicating poor quality and 5 indicating excellent

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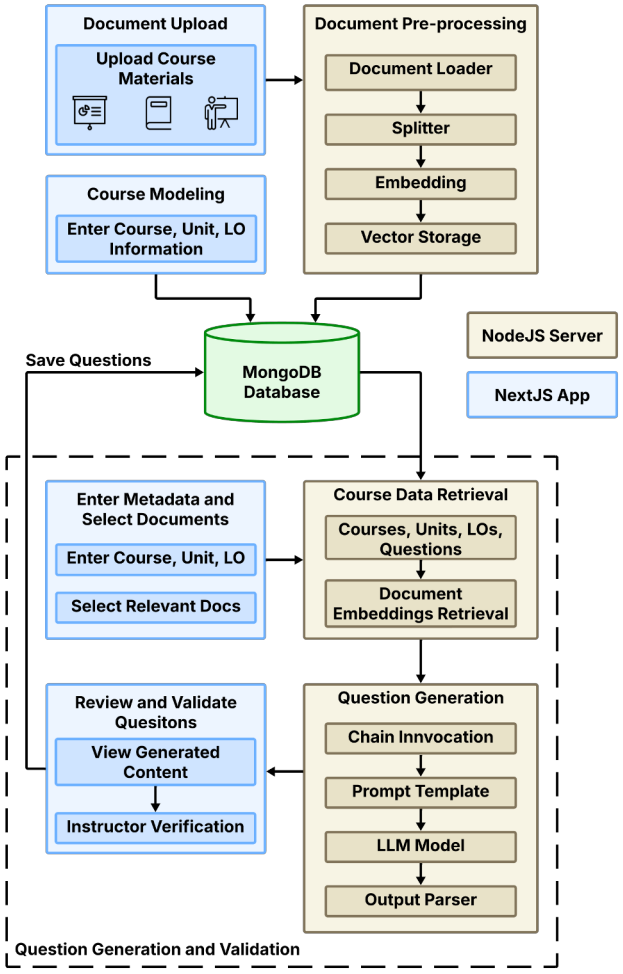


Fig. 2. System architecture for question generation with client-server framework.

quality. This rating was designed to serve as a holistic measure of each question's effectiveness, considering factors such as relevance, clarity, and alignment with the specified LOs.

V. Results and Discussion

A. Analysis of Instructor Validation Ratings

An analysis of instructor evaluations, summarized in Table II, indicates that Method-2 achieved a slightly higher overall mean rating (3.400) compared to Method-1 (3.286) across all courses. However, the corresponding p-value from a one-tailed t-test ($p = 0.296$) exceeds the conventional significance level of $\alpha = 0.05$, indicating that the observed difference is not statistically significant. These findings suggest that, from the instructors' perspective, both methods produce MCQs and feedback of comparable quality.

Table II also presents results by course. Biology received the highest overall ratings, with Method-2 outperforming Method-1 by a noticeable margin (4.400 vs. 3.920); however, the

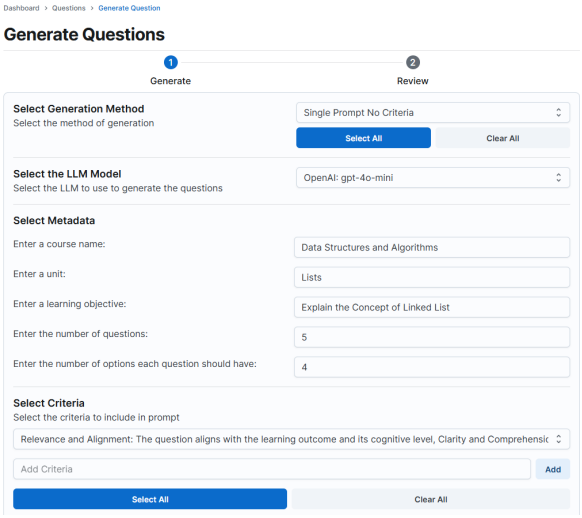


Fig. 3. Instructor interface for generating questions with different prompting methods, LLM models, and QVC.

difference was not statistically significant ($p = 0.076$). In contrast, Law and Communication received lower overall ratings, and in both cases, Method-2 performed slightly worse than Method-1 (2.933 vs. 3.033 and 2.667 vs. 2.733, respectively), with the differences also failing to reach statistical significance ($p = 0.363$ and $p = 0.423$, respectively). These patterns suggest possible disciplinary variation in how effectively LLM-generated questions meet instructor expectations, warranting further investigation.

B. Discussion

These findings have important implications for understanding the effectiveness of different techniques for embedding QVC in prompt engineering for LLMs.

First, the instructors across all three courses consistently rated the MCQs favorably, regardless of the generation method used, indicating that LLMs can produce reasonable formative assessment items using a zero-shot approach without the need for human feedback or embedded QVC. However, while both methods produced questions of similar quality, notable differences were observed across the three disciplines: Biology yielded the highest-rated questions, followed by Law, with

TABLE II
Summary Statistics of Instructor Validation Ratings

Course	Number of Questions per Method	Mean Rating (Method-1)	Mean Rating (Method-2)	p-value
Principles of Biology I	25	3.920	4.400	0.076
Administrative Law of Canada	30	3.033	2.933	0.363
Group and Organizational Communication	15	2.733	2.667	0.423
Total	70	3.286	3.400	0.296

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Communication rated the lowest. These differences may reflect the inherent variability in how well LLMs understand and generate content for different domains, or they could result from bias, as only the instructor of each course evaluated the questions. Further research involving multiple expert raters for each subject is needed to determine whether these domain differences are due to evaluator bias or a genuine limitation of LLM performance across subject areas.

Second, the lack of statistically significant differences between generation methods, with and without QVCs embedded in the prompts, suggests that QVCs alone do not provide sufficiently strong contextual guidance to meaningfully influence the quality of the generated content. These findings align with prior research indicating that, while LLMs can generate broadly acceptable questions, they often struggle to effectively incorporate instructional frameworks such as Bloom's Taxonomy [39], [52] and, in our case, QVC. This underscores the need for further research into methods for aligning LLM-generated educational content more closely with human expectations and pedagogical standards. Overall, this study contributes to the growing body of literature focused on identifying when prompt engineering and RAG techniques enhance model performance on educational tasks, and when their impact may be limited or even counterproductive.

In addition to the primary methods examined, this study also explored alternative approaches for embedding QVC into the prompt, including a two-round prompting method and a multi-round prompting method. In the two-round approach, the LLM first generates a specified number of questions based on the course, unit, and LO. Each question is then individually evaluated and revised by prompting the model with the question along with a list of QVC to ensure alignment with the specified criteria. In the multi-round prompting method, each QVC is assessed in isolation. The model generates a question and evaluates it against one criterion at a time, revising it accordingly before moving on to the next criterion. Instructors can control which criteria to apply and the order of evaluation.

These approaches are conceptually related to n-shot prompting methods. In this context, each additional round of prompting functions similarly to providing an example or refinement, potentially helping the model better internalize expectations. However, despite the increased structure and iteration provided by these multi-step methods, our initial findings from expert evaluation showed that they did not yield significantly different or improved results compared to Method-1 and Method-2. This suggests that increasing the number of refinement steps or embedded criteria may not necessarily enhance question quality in the absence of more robust instructional context or human feedback.

VI. Conclusion and Future Work

Leveraging the generative capabilities of LLMs has the potential to enhance the adaptive features of future ALSs, enabling more personalized and fine-tuned learning experiences. To address common challenges such as hallucinations in AI-generated content and to improve instructor efficiency, we proposed a QVC-based approach for generating pedagogically sound MCQs. This study examined whether embedding QVC into the prompting process could improve the quality of the generated questions. However, our results show no statistically significant difference in the quality of validated MCQs produced with or without QVC using a zero-shot prompting approach. These findings suggest that while LLMs demonstrate strong baseline performance, they may not fully interpret or apply the additional educational context in the way a human instructor would.

Although our findings showed that injecting QVC into prompts did not significantly improve question quality, the broader application of LLMs and the framework we have developed for MCQ generation remains highly encouraging. Even in the absence of structured guidance, such as QVC, our framework demonstrated the ability to produce reasonably high-quality assessment items efficiently. These methods can support future experimental designs aimed at optimizing LLM output for educational content generation.

Future work should focus on refining the formulation of QVC and developing more nuanced evaluation metrics that can capture subtle variations in question quality. Alternative approaches, such as multi-shot prompting, should also be explored in greater depth to assess their potential for improving question quality and contextual relevance. Beyond prompt engineering and RAG, alternative strategies for improving LLM output in educational contexts warrant exploration. Approaches such as model fine-tuning [53], domain-specific pretraining [54], reinforcement learning with human feedback (RLHF) [55], curriculum and domain modeling [56]–[58], Graph RAG [59], and knowledge graph-based representations have shown particular promise [60]–[62]. Furthermore, as in other domains, experimental reproducibility is a critical concern in education and remains a key limitation of proprietary models [63]. To mitigate this, further research should investigate the use of open-source models and more interpretable models to reduce potential biases and limitations introduced by commercial LLMs, ensuring greater transparency, adaptability, and alignment with educational goals [64], [65].

To ensure broader applicability, future work should involve multiple domain experts across more subject areas than explored here to validate the quality and pedagogical relevance of the generated questions. Expanding the study to include a larger question bank and a broader range of disciplines will be essential for assessing the generalizability of the findings. Additionally, investigating the long-term impact of AI-generated educational content on learner outcomes and engagement will be crucial for evaluating the real-world effectiveness and sustainability of these approaches in instructional settings.

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REFERENCES

[1] D. Nguyen, L. Howard, H. Yan, and F. Lin, "Adaptive sequence learning: Contextual multi-armed bandit approach," in *2023 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDCom/CyberSciTech)*. IEEE, Nov. 2023, p. 0751–0756. [Online]. Available: http://dx.doi.org/10.1109/DASC/PiCom/CBDCom/Cy59711.2023.10361498

[2] L. M. D. Nguyen, F. Lin, and M. Chang, *Generating Learning Sequences Using Contextual Bandit Algorithms*. Springer Nature Switzerland, 2024, p. 320–329. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-63028-6_26

[3] Q. U. Ain, M. A. Chatti, P. A. Meteng Kamdem, R. Alatrash, S. Joarder, and C. Siepmann, "Learner modeling and recommendation of learning resources using personal knowledge graphs," in *Proceedings of the 14th Learning Analytics and Knowledge Conference*, ser. LAK '24. ACM, Mar. 2024, p. 273–283. [Online]. Available: http://dx.doi.org/10.1145/3636555.3636881

[4] E. Haghshenas, A. Mazaheri, A. Gholipour, M. Tavakoli, N. Zandi, H. Narimani *et al.*, "Introducing a new intelligent adaptive learning content generation method," in *The Second International Conference on E-Learning and E-Teaching (ICELET 2010)*. IEEE, Dec. 2010, p. 65–71. [Online]. Available: http://dx.doi.org/10.1109/ICELET.2010.5708382

[5] T. C. Reeves, J. Herrington, and R. Oliver, "Authentic activities and online learning," in *HERDSA 2002 quality conversations*. Higher Education Research and Development Society of Australasia, Inc, 2002.

[6] V. A. Nguyen, "The impact of online learning activities on student learning outcome in blended learning course," *Journal of Information amp; Knowledge Management*, vol. 16, no. 04, p. 1750040, Nov. 2017. [Online]. Available: http://dx.doi.org/10.1142/S021964921750040X

[7] F. Lin and S. De Silva, "An approach to generating adaptive feedback for online formative assessment," in *Augmented Intelligence and Intelligent Tutoring Systems*. Springer Nature Switzerland, 2023, p. 88–99. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-32883-1_8

[8] R. Morland, L. Wang, and F. Lin, *Fast Weakness Identification for Adaptive Feedback*. Springer Nature Switzerland, 2024, p. 40–54. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-63028-6_4

[9] M. Cukurova, X. Miao, and R. Brooker, *Adoption of Artificial Intelligence in Schools: Unveiling Factors Influencing Teachers' Engagement*. Springer Nature Switzerland, 2023, p. 151–163. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-36272-9_13

[10] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal *et al.*, "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., 2020, pp. 1877–1901. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf

[11] OpenAI, J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya *et al.*, "Gpt-4 technical report," 2024. [Online]. Available: https://arxiv.org/abs/2303.08774

[12] N. F. Davar, M. A. A. Dewan, and X. Zhang, "Ai chatbots in education: Challenges and opportunities," *Information*, vol. 16, no. 3, p. 235, Mar. 2025. [Online]. Available: http://dx.doi.org/10.3390/info16030235

[13] J. L. Steele, "To gpt or not gpt? empowering our students to learn with ai," *Computers and Education: Artificial Intelligence*, vol. 5, p. 100160, 2023. [Online]. Available: http://dx.doi.org/10.1016/j.caeai.2023.100160

[14] X. Yuan, T. Wang, Y.-H. Wang, E. Fine, R. Abdelghani, H. Sauzéon, and P.-Y. Oudeyer, "Selecting better samples from pre-trained LLMs: A case study on question generation," in *Findings of the Association for Computational Linguistics: ACL 2023*, A. Rogers, J. Boyd-Graber, and N. Okazaki, Eds. Toronto, Canada: Association for Computational Linguistics, Jul. 2023, pp. 12 952–12 965. [Online]. Available: https://aclanthology.org/2023.findings-acl.820/

[15] E. Elina, M. Maimunah, and Y. Roza, "Analysis of the ability of smp/mts mathematics teachers in making akm type questions," *Jurnal Gantang*, vol. 7, no. 1, p. 47–57, Aug. 2022. [Online]. Available: http://dx.doi.org/10.31629/jg.v7i1.4466

[16] S. K. Donkor, L. G. Nyavor, P. Addai-Tuffour, R. S. K. Abieraba, R. S. Sorkpor, and R. Ocansey, "Intra-observer evaluation of teacher-student questioning behaviours: A case study of the quality and frequency of questions administered in physical education theory class," *International Journal of Physical Education, Sports and Health*, vol. 8, no. 5, p. 192–197, Sep. 2021. [Online]. Available: http://dx.doi.org/10.22271/kheljournal.2021.v8.i5c.2286

[17] F. Miao and W. Holmes, *Guidance for generative AI in education and research*. UNESCO, 2023.

[18] M. Gottlieb, J. Bailitz, M. Fix, E. Shappell, and M. J. Wagner, "Educator's blueprint: A how-to guide for developing high-quality multiple-choice questions," *AEM Education and Training*, vol. 7, no. 1, Jan. 2023. [Online]. Available: http://dx.doi.org/10.1002/aet2.10836

[19] E. J. Palmer and P. G. Devitt, "Assessment of higher order cognitive skills in undergraduate education: modified essay or multiple choice questions? research paper," *BMC Medical Education*, vol. 7, no. 1, Nov. 2007. [Online]. Available: http://dx.doi.org/10.1186/1472-6920-7-49

[20] S. S. Mucciaccia, T. Meireles Paixão, F. Wall Mutz, C. Santos Badue, A. Ferreira de Souza, and T. Oliveira-Santos, "Automatic multiple-choice question generation and evaluation systems based on LLM: A study case with university resolutions," in *Proceedings of the 31st International Conference on Computational Linguistics*, O. Rambow, L. Wanner, M. Apidianaki, H. Al-Khalifa, B. D. Eugenio, and S. Schockaert, Eds. Abu Dhabi, UAE: Association for Computational Linguistics, Jan. 2025, pp. 2246–2260. [Online]. Available: https://aclanthology.org/2025.coling-main.154/

[21] T. M. Haladyna and S. M. Downing, "A taxonomy of multiple-choice item-writing rules," *Applied Measurement in Education*, vol. 2, no. 1, p. 37–50, Jan. 1989. [Online]. Available: http://dx.doi.org/10.1207/s15324818ame0201_3

[22] M. Tarrant, A. Knierim, S. K. Hayes, and J. Ware, "The frequency of item writing flaws in multiple-choice questions used in high stakes nursing assessments," *Nurse Education Today*, vol. 26, no. 8, p. 662–671, Dec. 2006. [Online]. Available: http://dx.doi.org/10.1016/j.nedt.2006.07.006

[23] D. DiBattista, "Making the most of multiple-choice questions: Getting beyond remembering," *Collected Essays on Learning and Teaching*, vol. 1, p. 119–122, Jan. 2008. [Online]. Available: http://dx.doi.org/10.22329/celt.v1i0.3190

[24] M. Heilman and N. A. Smith, "Good question! statistical ranking for question generation," in *Human Language Technologies: The 2010 Annual Conference of the North American Chapter of the Association for Computational Linguistics*, R. Kaplan, J. Burstein, M. Harper, and G. Penn, Eds. Los Angeles, California: Association for Computational Linguistics, Jun. 2010, pp. 609–617. [Online]. Available: https://aclanthology.org/N10-1086/

[25] N. Mulla and P. Gharpure, "Automatic question generation: a review of methodologies, datasets, evaluation metrics, and applications," *Progress in Artificial Intelligence*, vol. 12, no. 1, p. 1–32, Jan. 2023. [Online]. Available: http://dx.doi.org/10.1007/s13748-023-00295-9

[26] J. Amidei, P. Piwek, and A. Willis, "Evaluation methodologies in automatic question generation 2013-2018," in *Proceedings of the 11th International Conference on Natural Language Generation*, E. Krahmer, A. Gatt, and M. Goudbeek, Eds. Tilburg University, The Netherlands: Association for Computational Linguistics, Nov. 2018, pp. 307–317. [Online]. Available: https://aclanthology.org/W18-6537/

[27] B. Das, M. Majumder, S. Phadikar, and A. A. Sekh, "Automatic question generation and answer assessment: a survey," *Research and Practice in Technology Enhanced Learning*, vol. 16, no. 1, Mar. 2021. [Online]. Available: http://dx.doi.org/10.1186/s41039-021-00151-1

[28] X. Du, J. Shao, and C. Cardie, "Learning to ask: Neural question generation for reading comprehension," in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, R. Barzilay and M.-Y. Kan, Eds. Vancouver, Canada: Association for Computational Linguistics, Jul. 2017, pp. 1342–1352. [Online]. Available: https://aclanthology.org/P17-1123/

[29] G. Kurdi, J. Leo, B. Parsia, U. Sattler, and S. Al-Emari, "A systematic review of automatic question generation for educational purposes," *International Journal of Artificial Intelligence in Education*, vol. 30, no. 1, p. 121–204, Nov. 2019. [Online]. Available: http://dx.doi.org/10.1007/s40593-019-00186-y

[30] Y.-A. Bachiri and H. Mouncif, "Artificial intelligence system in aid of pedagogical engineering for knowledge assessment on mooc platforms: Open edx and moodle," *International Journal of Emerging Technologies in Learning (iJET)*, vol. 18, no. 05, p. 144–160, Mar. 2023. [Online]. Available: http://dx.doi.org/10.3991/ijet.v18i05.36589

[31] F. Akyön, D. Çavuşoğlu, C. Cengiz, S. O. Altinuç, and A. Temizel, "Automated question generation and question answering from Turkish

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texts," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 30, no. 5, pp. 1931–1940, Jul. 2022.

[32] P. Chomphooyod, A. Suchato, N. Tuaycharoen, and P. Punyabukkana, "English grammar multiple-choice question generation using text-to-text transfer transformer," *Computers and Education: Artificial Intelligence*, vol. 5, p. 100158, 2023. [Online]. Available: http://dx.doi.org/10.1016/j.caeai.2023.100158

[33] L. Wen, Z. Zhang, and W. Zhao, "Generating controllable questions from knowledge graph via SPARQL encoding and reinforcement learning," in *Lecture Notes in Computer Science*, ser. Lecture notes in computer science. Singapore: Springer Nature Singapore, 2024, pp. 475–487.

[34] G. Jhajj and Y. Nomura, "Jack and the beansTALK: Towards question answering in plant biology," in *Eighth Widening NLP Workshop (WiNLP 2024) Phase II*, 2024. [Online]. Available: https://openreview.net/forum?id=0DlJEPHHKe

[35] C. H. Chen and M. F. Shiu, "Kaqg: A knowledge-graph-enhanced rag for difficulty-controlled question generation," May 2025. [Online]. Available: http://dx.doi.org/10.36227/techrxiv.174681425.54614303/v1

[36] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., 2020, pp. 9459–9474. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf

[37] G. Gorgun and O. Bulut, "Exploring quality criteria and evaluation methods in automated question generation: A comprehensive survey," *Education and Information Technologies*, vol. 29, no. 18, p. 24111–24142, Jun. 2024. [Online]. Available: http://dx.doi.org/10.1007/s10639-024-12771-3

[38] V. Raina and M. Gales, "Multiple-choice question generation: Towards an automated assessment framework," 2022. [Online]. Available: https://arxiv.org/abs/2209.11830

[39] S. Maity, A. Deroy, and S. Sarkar, "How effective is gpt-4 turbo in generating school-level questions from textbooks based on bloom's revised taxonomy?" 2024. [Online]. Available: https://arxiv.org/abs/2406.15211

[40] P. G. Herrero and A. R. Medina, "The cognitive requirement in the multiple-choice exams in the university context and its relationship with learning approaches, self-regulation, teaching methods and academic performance," *Eur. J. Investig. Health Psychol. Educ.*, vol. 9, no. 3, pp. 177–187, Sep. 2019.

[41] G. S. Frandsen and M. I. Schwartzbach, "A singular choice for multiple choice," *ACM SIGCSE Bulletin*, vol. 38, no. 4, p. 34–38, Jun. 2006. [Online]. Available: http://dx.doi.org/10.1145/1189136.1189164

[42] A. Sayapin, "Multiple choice assessments: Evaluation of quality," in *2013 IEEE Global Engineering Education Conference (EDUCON)*. IEEE, Mar. 2013, p. 352–355. [Online]. Available: http://dx.doi.org/10.1109/EduCon.2013.6530128

[43] F. Lin and S. De Silva, *An Approach to Generating Adaptive Feedback for Online Formative Assessment*. Springer Nature Switzerland, 2023, p. 88–99. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-32883-1_8

[44] A. M. Olney, "Generating multiple choice questions from a textbook: Llms match human performance on most metrics." *Grantee Submission*, 2023.

[45] M. R. Kabir and F. Lin, "An llm-powered adaptive practicing system." in *LLM@ AIED*, 2023, pp. 43–52.

[46] F. Lin, R. Morland, and H. Yan, "Quizmaster: An adaptive formative assessment system," in *Generative Intelligence and Intelligent Tutoring Systems*, A. Sifaleras and F. Lin, Eds. Cham: Springer Nature Switzerland, 2024, pp. 55–67. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-63028-6_5

[47] G. Jhajj and F. Lin, "Augmenting japanese language acquisition via llms and asr," in *IEEE Smart World Congress 2025 (IEEE SWC'25)*, Calgary, Canada, Aug. 2025, p. 3.84.

[48] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, "Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing," *ACM Comput. Surv.*, vol. 55, no. 9, Jan. 2023. [Online]. Available: https://doi.org/10.1145/3560815

[49] L. Wang, R. Song, W. Guo, and H. Yang, "Exploring prompt pattern for generative artificial intelligence in automatic question generation," *Interactive Learning Environments*, vol. 33, no. 3, p. 2559–2584, Oct. 2024. [Online]. Available: http://dx.doi.org/10.1080/10494820.2024.2412082

[50] H. Chase, "LangChain," Oct. 2022. [Online]. Available: https://github.com/langchain-ai/langchain

[51] H. Yin, A. Aryani, and N. Nambiar, "Evaluating the performance of large language models for sdg mapping (technical report)," 2024. [Online]. Available: https://arxiv.org/abs/2408.02201

[52] K. Hwang, K. Wang, M. Alomair, F.-S. Choa, and L. K. Chen, "Towards automated multiple choice question generation and evaluation: Aligning with bloom's taxonomy," in *Artificial Intelligence in Education*. Springer Nature Switzerland, 2024, p. 389–396. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-64299-9_35

[53] J. Ling and M. Afzaal, "Automatic question-answer pairs generation using pre-trained large language models in higher education," *Computers and Education: Artificial Intelligence*, vol. 6, p. 100252, Jun. 2024. [Online]. Available: http://dx.doi.org/10.1016/j.caeai.2024.100252

[54] F. Zhang, C. Li, O. Henkel, W. Xing, S. Baral, N. Heffernan, and H. Li, "Math-llms: Ai cyberinfrastructure with pre-trained transformers for math education," *International Journal of Artificial Intelligence in Education*, vol. 35, no. 2, p. 509–532, Jul. 2024. [Online]. Available: http://dx.doi.org/10.1007/s40593-024-00416-y

[55] A. Scarlatos, D. Smith, S. Woodhead, and A. Lan, *Improving the Validity of Automatically Generated Feedback via Reinforcement Learning*. Springer Nature Switzerland, 2024, p. 280–294. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-64302-6_20

[56] F. Lin and R. Morland, *Curriculum Modeling for Adaptive Learning*. Springer Nature Switzerland, 2025, p. 57–72. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-92970-0_5

[57] H. Abu-Rasheed, C. Jumbo, R. Al Amin, C. Weber, V. Wiese, R. Obermaisser, and M. Fathi, "Llm-assisted knowledge graph completion for curriculum and domain modelling in personalized higher education recommendations," in *2025 IEEE Global Engineering Education Conference (EDUCON)*. IEEE, Apr. 2025, p. 1–5. [Online]. Available: http://dx.doi.org/10.1109/EDUCON62633.2025.11016377

[58] Q. U. Ain, M. A. Chatti, A. Shakhshir, J. Qussa, R. Alatrash, and S. Joarder, "Top-down vs. bottom-up approaches for automatic educational knowledge graph construction in coursemapper," 2025. [Online]. Available: https://arxiv.org/abs/2505.10069

[59] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody *et al.*, "From local to global: A graph rag approach to query-focused summarization," 2025. [Online]. Available: https://arxiv.org/abs/2404.16130

[60] G. Jhajj, X. Zhang, J. R. Gustafson, F. Lin, and M. P.-C. Lin, *Educational Knowledge Graph Creation and Augmentation via LLMs*. Springer Nature Switzerland, 2024, p. 292–304. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-63031-6_25

[61] J. R. D. Gustafson, G. Jhajj, X. Zhang, and F. O. Lin, "Enhancing project-based learning with a genai tool based on retrieval: Augmented generation and knowledge graphs," in *AI Applications and Strategies in Teacher Education*. IGI Global, 2025, pp. 161–194. [Online]. Available: http://dx.doi.org/10.4018/979-8-3693-5443-8.ch006

[62] J. R. D. Gustafson, X. Zhang, G. Jhajj, and F. Lin, "Representing and tracing students' cognitive processes in Project-Based learning through the Function-Behavior-Structure framework and knowledge graphs," in *IEEE Smart World Congress 2025 (IEEE SWC'25)*, Calgary, Canada, Aug. 2025, p. 6.56.

[63] É. Ollion, R. Shen, A. Macanovic, and A. Chatelain, "The dangers of using proprietary llms for research," *Nature Machine Intelligence*, vol. 6, no. 1, p. 4–5, Jan. 2024. [Online]. Available: http://dx.doi.org/10.1038/s42256-023-00783-6

[64] M. P.-C. Lin, D. Chang, S. Hall, and G. Jhajj, *Preliminary Systematic Review of Open-Source Large Language Models in Education*. Springer Nature Switzerland, 2024, p. 68–77. [Online]. Available: http://dx.doi.org/10.1007/978-3-031-63028-6_6

[65] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, p. 206–215, May 2019. [Online]. Available: http://dx.doi.org/10.1038/s42256-019-0048-x

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